

ECEN 629 Convex Optimization for Electrical Engineering

Chapter 5. Duality

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Lagrangian

standard form problem (not necessarily convex)

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

variable $x \in \mathbf{R}^n$, domain $\mathcal{D}$, optimal value p^*

Lagrangian: $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \rightarrow \mathbf{R}$, with $\text{dom } L = \mathcal{D} \times \mathbf{R}^m \times \mathbf{R}^p$,

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x)$$

- weighted sum of objective and constraint functions
- λ_i is Lagrange multiplier associated with $f_i(x) \leq 0$
- ν_i is Lagrange multiplier associated with $h_i(x) = 0$

Lagrange dual function

Lagrange dual function: $g : \mathbf{R}^m \times \mathbf{R}^p \rightarrow \mathbf{R}$,

$$\begin{aligned} g(\lambda, \nu) &= \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \\ &= \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x) \right) \end{aligned}$$

g is concave, can be $-\infty$ for some λ, ν

lower bound property: if $\lambda \succeq 0$, then $g(\lambda, \nu) \leq p^*$

proof: if $\tilde{x}$ is feasible and $\lambda \succeq 0$, then

$$f_0(\tilde{x}) \geq L(\tilde{x}, \lambda, \nu) \geq \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = g(\lambda, \nu)$$

minimizing over all feasible $\tilde{x}$ gives $p^* \geq g(\lambda, \nu)$

Least-norm solution of linear equations

$$\begin{array}{ll}\text{minimize} & x^T x \\ \text{subject to} & Ax = b\end{array}$$

dual function

- Lagrangian is $L(x, \nu) = x^T x + \nu^T (Ax - b)$
- to minimize L over x , set gradient equal to zero:

$$\nabla_x L(x, \nu) = 2x + A^T \nu = 0 \quad \implies \quad x = -(1/2)A^T \nu$$

- plug in in L to obtain g :

$$g(\nu) = L((-1/2)A^T \nu, \nu) = -\frac{1}{4}\nu^T A A^T \nu - b^T \nu$$

a concave function of ν

lower bound property: $p^* \geq -(1/4)\nu^T A A^T \nu - b^T \nu$ for all ν

Standard form LP

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \succeq 0\end{array}$$

dual function

- Lagrangian is

$$\begin{aligned}L(x, \lambda, \nu) &= c^T x + \nu^T (Ax - b) - \lambda^T x \\ &= -b^T \nu + (c + A^T \nu - \lambda)^T x\end{aligned}$$

- L is affine in x , hence

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \begin{cases} -b^T \nu & A^T \nu - \lambda + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

g is linear on affine domain $\{(\lambda, \nu) \mid A^T \nu - \lambda + c = 0\}$, hence concave

lower bound property: $p^* \geq -b^T \nu$ if $A^T \nu + c \succeq 0$

Equality constrained norm minimization

$$\begin{array}{ll}\text{minimize} & \|x\| \\ \text{subject to} & Ax = b\end{array}$$

dual function

$$g(\nu) = \inf_x (\|x\| - \nu^T Ax + b^T \nu) = \begin{cases} b^T \nu & \|A^T \nu\|_* \leq 1 \\ -\infty & \text{otherwise} \end{cases}$$

where $\|v\|_* = \sup_{\|u\| \leq 1} u^T v$ is dual norm of $\|\cdot\|$

proof: follows from $\inf_x (\|x\| - y^T x) = 0$ if $\|y\|_* \leq 1$, $-\infty$ otherwise

- if $\|y\|_* \leq 1$, then $\|x\| - y^T x \geq 0$ for all x , with equality if $x = 0$
- if $\|y\|_* > 1$, choose $x = tu$ where $\|u\| \leq 1$, $u^T y = \|y\|_* > 1$:

$$\|x\| - y^T x = t(\|u\| - \|y\|_*) \rightarrow -\infty \quad \text{as } t \rightarrow \infty$$

lower bound property: $p^* \geq b^T \nu$ if $\|A^T \nu\|_* \leq 1$

Two-way partitioning

$$\begin{array}{ll}\text{minimize} & x^T W x \\ \text{subject to} & x_i^2 = 1, \quad i = 1, \dots, n\end{array}$$

- a nonconvex problem; feasible set contains 2^n discrete points
- interpretation: partition $\{1, \dots, n\}$ in two sets; W_{ij} is cost of assigning i, j to the same set; $-W_{ij}$ is cost of assigning to different sets

dual function

$$\begin{aligned} g(\nu) &= \inf_x (x^T W x + \sum_i \nu_i (x_i^2 - 1)) = \inf_x x^T (W + \mathbf{diag}(\nu)) x - \mathbf{1}^T \nu \\ &= \begin{cases} -\mathbf{1}^T \nu & W + \mathbf{diag}(\nu) \succeq 0 \\ -\infty & \text{otherwise} \end{cases} \end{aligned}$$

lower bound property: $p^* \geq -\mathbf{1}^T \nu$ if $W + \mathbf{diag}(\nu) \succeq 0$

example: $\nu = -\lambda_{\min}(W)\mathbf{1}$ gives bound $p^* \geq n\lambda_{\min}(W)$

Lagrange dual and conjugate function

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & Ax \preceq b, \quad Cx = d\end{array}$$

dual function

$$\begin{aligned}g(\lambda, \nu) &= \inf_{x \in \text{dom } f_0} (f_0(x) + (A^T \lambda + C^T \nu)^T x - b^T \lambda - d^T \nu) \\ &= -f_0^*(-A^T \lambda - C^T \nu) - b^T \lambda - d^T \nu\end{aligned}$$

- recall definition of conjugate $f^*(y) = \sup_{x \in \text{dom } f} (y^T x - f(x))$
- simplifies derivation of dual if conjugate of f_0 is known

example: entropy maximization

$$f_0(x) = \sum_{i=1}^n x_i \log x_i, \quad f_0^*(y) = \sum_{i=1}^n e^{y_i - 1}$$

The dual problem

Lagrange dual problem

$$\begin{array}{ll}\text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0\end{array}$$

- finds best lower bound on p^* , obtained from Lagrange dual function
- a convex optimization problem; optimal value denoted d^*
- λ, ν are dual feasible if $\lambda \succeq 0, (\lambda, \nu) \in \mathbf{dom} g$
- often simplified by making implicit constraint $(\lambda, \nu) \in \mathbf{dom} g$ explicit

example: standard form LP and its dual (page 5–5)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \succeq 0\end{array}$$

$$\begin{array}{ll}\text{maximize} & -b^T \nu \\ \text{subject to} & A^T \nu + c \succeq 0\end{array}$$

Weak and strong duality

weak duality: $d^* \leq p^*$

- always holds (for convex and nonconvex problems)
- can be used to find nontrivial lower bounds for difficult problems
for example, solving the SDP

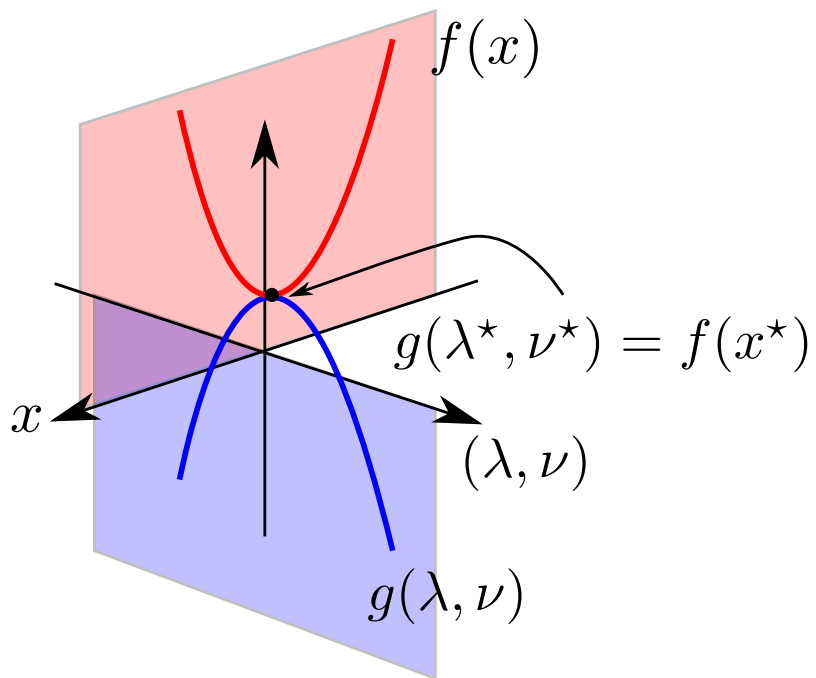
$$\begin{array}{ll}\text{maximize} & -\mathbf{1}^T \nu \\ \text{subject to} & W + \mathbf{diag}(\nu) \succeq 0\end{array}$$

gives a lower bound for the two-way partitioning problem on page 5–7

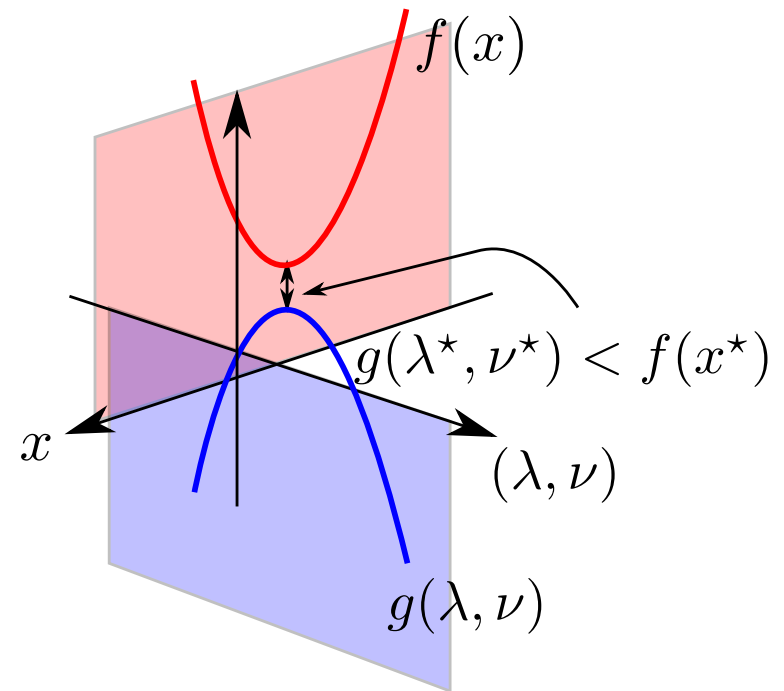
strong duality: $d^* = p^*$

- does not hold in general
- (usually) holds for convex problems
- conditions that guarantee strong duality in convex problems are called **constraint qualifications**

Duality and Duality Gap



strong duality



weak duality

Exercise 1

Consider the simple optimization problem:

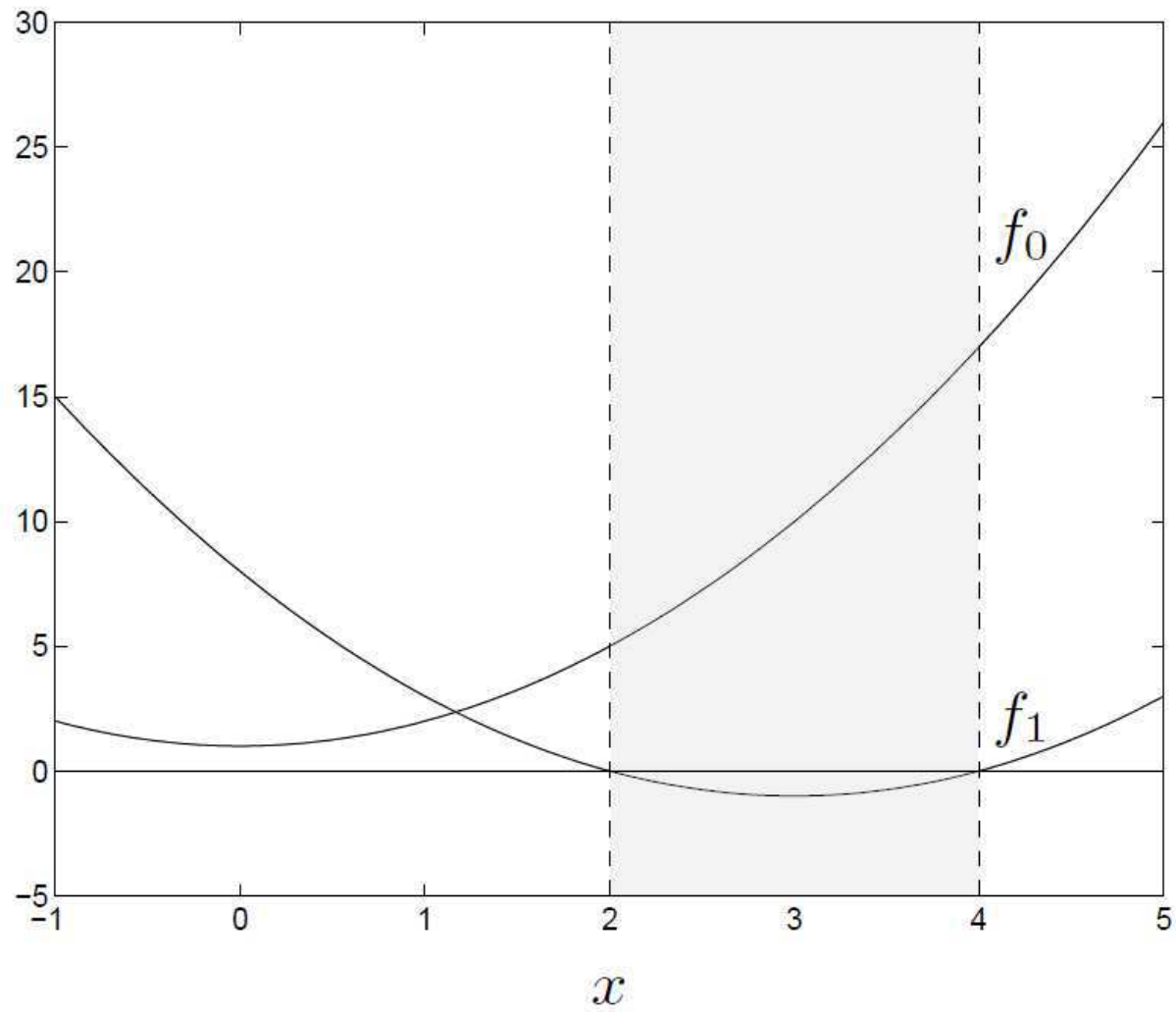
$$\begin{aligned} &\text{minimize: } x^2 + 1 \\ &\text{subject to: } (x - 2)(x - 4) \leq 0 \end{aligned}$$

1. Plot the functions. What are the feasible set, the optimal value, and the optimal solution?
2. Plot the Lagrangian $L(x, \lambda)$ for $\lambda = 1$ and $\lambda = 2$.
3. Find the Lagrange dual problem. What is the dual problem?
4. Verify the dual problem is concave, and find the dual optimal solution λ^* .
5. Let $p^*(u)$ be the optimal value of the problem

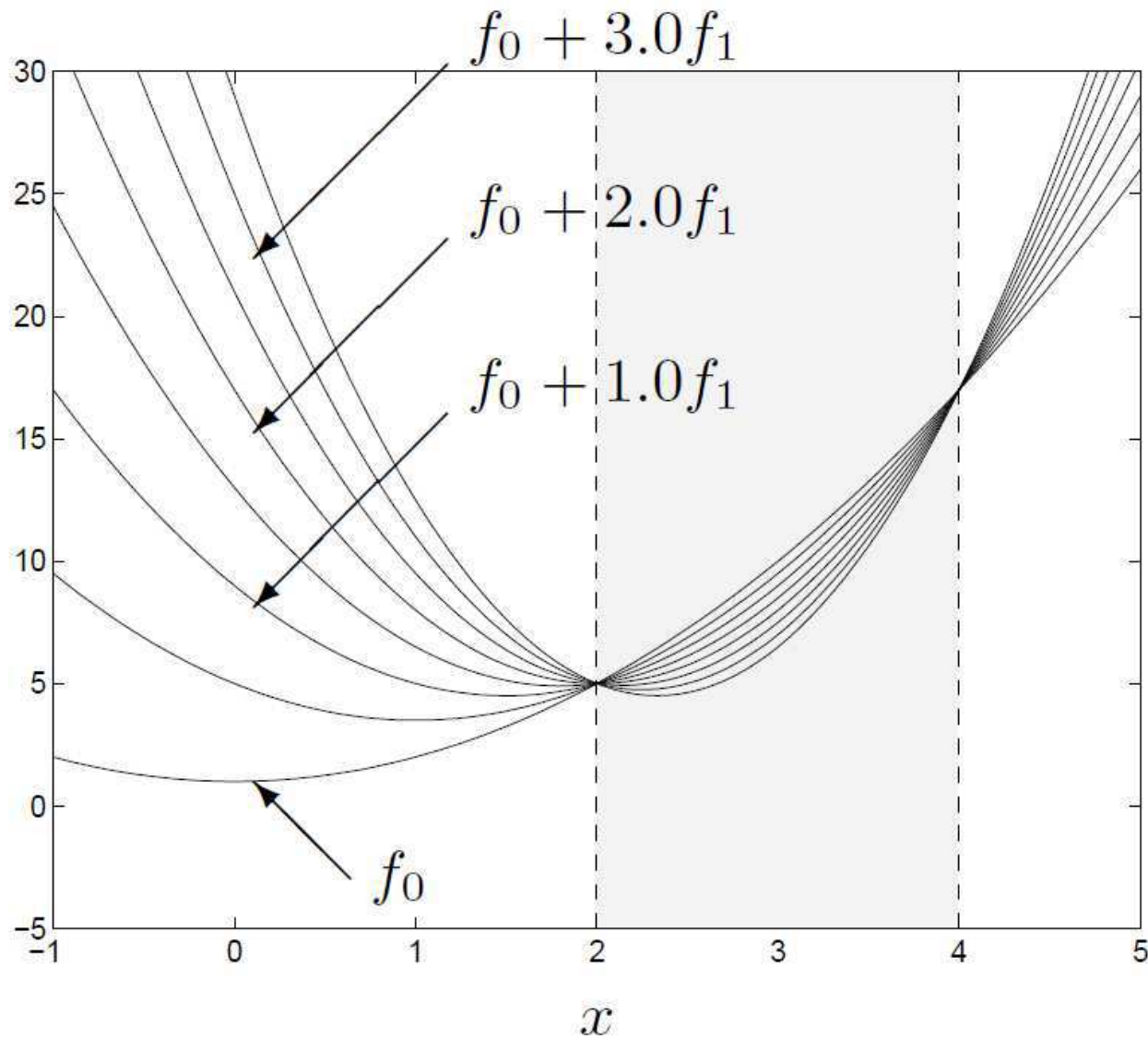
$$\begin{aligned} &\text{minimize: } x^2 + 1 \\ &\text{subject to: } (x - 2)(x - 4) \leq u \end{aligned}$$

as a function of the parameter u . Compute $\partial p^*(0)/\partial u$.

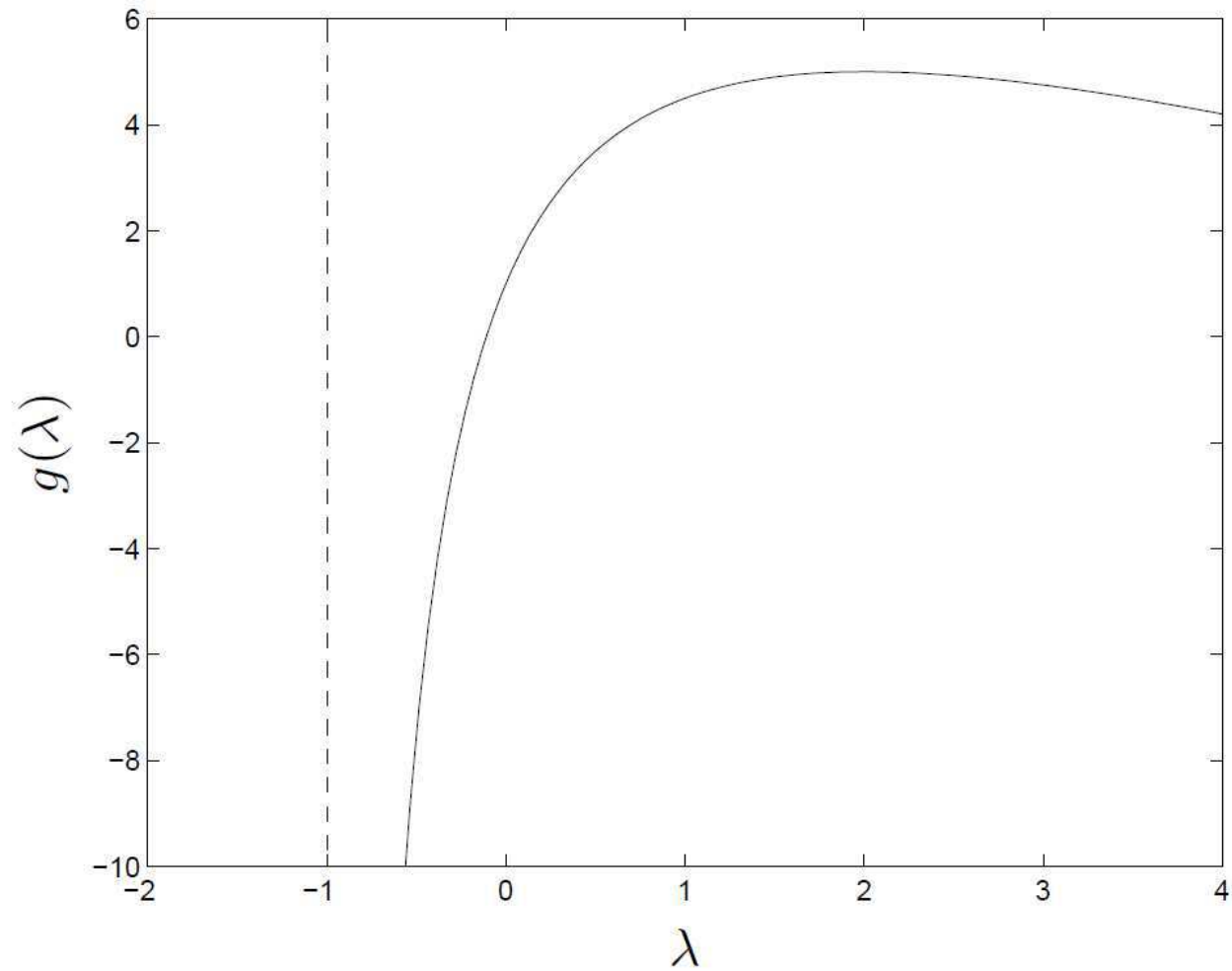
Exercise 1: Solutions



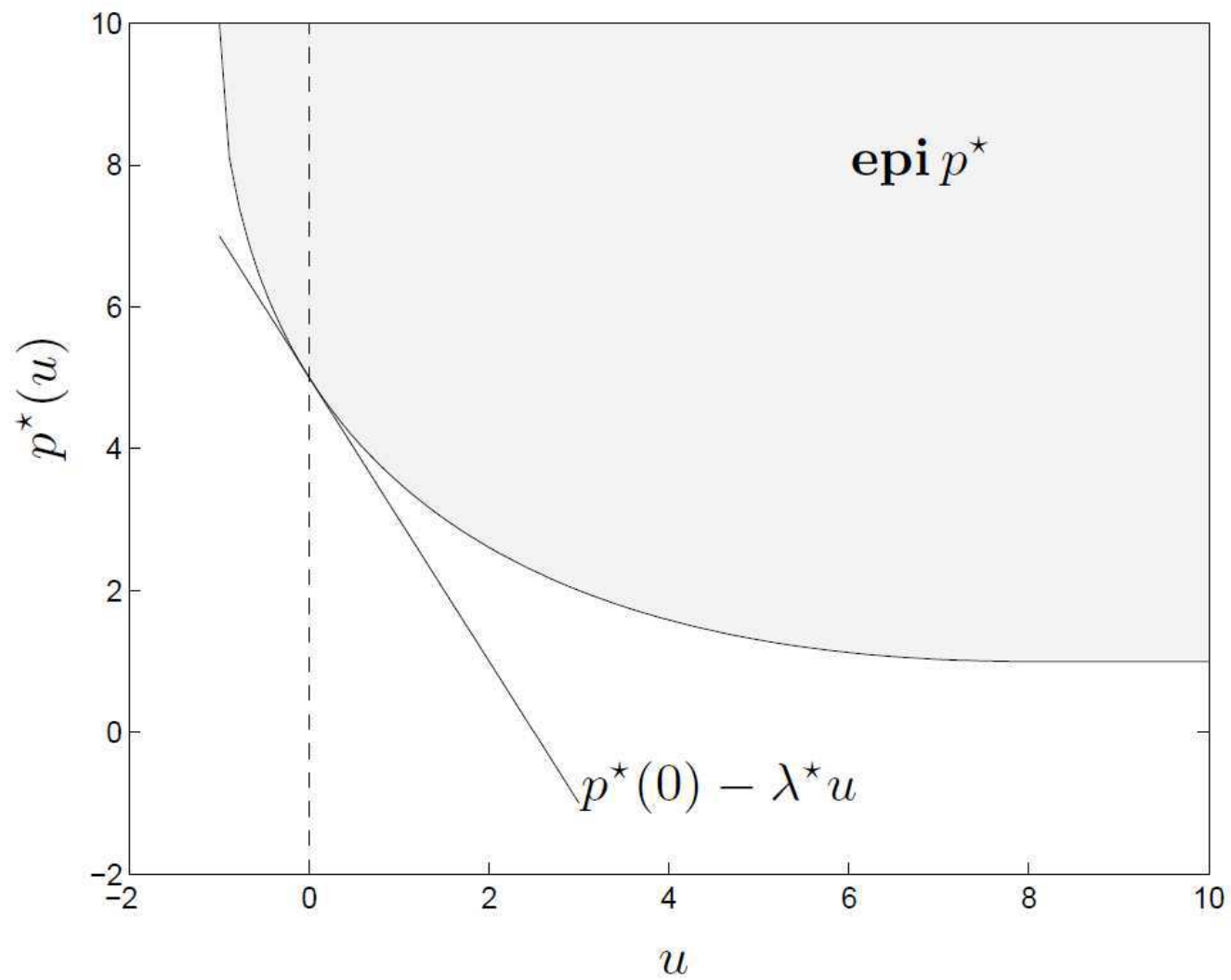
Exercise 1: Solutions



Exercise 1: Solutions



Exercise 1: Solutions



Slater's constraint qualification

strong duality holds for a convex problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

if it is strictly feasible, *i.e.*,

$$\exists x \in \mathbf{int} \mathcal{D} : \quad f_i(x) < 0, \quad i = 1, \dots, m, \quad Ax = b$$

- also guarantees that the dual optimum is attained (if $p^* > -\infty$)
- can be sharpened: *e.g.*, can replace $\mathbf{int} \mathcal{D}$ with $\mathbf{relint} \mathcal{D}$ (interior relative to affine hull); linear inequalities do not need to hold with strict inequality, . . .
- there exist many other types of constraint qualifications

Inequality form LP

primal problem

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax \preceq b\end{array}$$

dual function

$$g(\lambda) = \inf_x ((c + A^T \lambda)^T x - b^T \lambda) = \begin{cases} -b^T \lambda & A^T \lambda + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

dual problem

$$\begin{array}{ll}\text{maximize} & -b^T \lambda \\ \text{subject to} & A^T \lambda + c = 0, \quad \lambda \succeq 0\end{array}$$

- from Slater's condition: $p^* = d^*$ if $A\tilde{x} \prec b$ for some $\tilde{x}$
- in fact, $p^* = d^*$ except when primal and dual are infeasible

Infeasible LP

It is indeed possible that neither primal nor dual is feasible:

$$\begin{aligned} &\text{minimize: } x_2 - 2x_1 \\ &\text{subject to: } x_1 - x_2 \leq 1 \\ &\quad \quad \quad -x_1 + x_2 \leq -2 \\ &\quad \quad \quad x_1, x_2 \geq 0. \end{aligned}$$

What is the dual problem

Infeasible LP

It is indeed possible that neither primal nor dual is feasible:

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What is the dual problem

$$\begin{aligned} &\text{maximize: } -\lambda_1 + 2\lambda_2 \\ &\text{subject to: } \lambda_1 - \lambda_2 - \lambda_3 = 2 \\ &\quad \quad \quad -\lambda_1 + \lambda_2 - \lambda_4 = -1 \\ &\quad \quad \quad \lambda_1, \lambda_2, \lambda_3, \lambda_4 \geq 0. \end{aligned}$$

Why is this infeasible?

Quadratic program

primal problem (assume $P \in \mathbf{S}_{++}^n$)

$$\begin{array}{ll}\text{minimize} & x^T P x \\ \text{subject to} & Ax \preceq b\end{array}$$

dual function

$$g(\lambda) = \inf_x (x^T P x + \lambda^T (Ax - b)) = -\frac{1}{4} \lambda^T A P^{-1} A^T \lambda - b^T \lambda$$

dual problem

$$\begin{array}{ll}\text{maximize} & -(1/4) \lambda^T A P^{-1} A^T \lambda - b^T \lambda \\ \text{subject to} & \lambda \succeq 0\end{array}$$

- from Slater's condition: $p^* = d^*$ if $A\tilde{x} \prec b$ for some $\tilde{x}$
- in fact, $p^* = d^*$ always

A nonconvex problem with strong duality

$$\begin{array}{ll}\text{minimize} & x^T A x + 2b^T x \\ \text{subject to} & x^T x \leq 1\end{array}$$

$A \not\succeq 0$, hence nonconvex

dual function: $g(\lambda) = \inf_x (x^T (A + \lambda I)x + 2b^T x - \lambda)$

- unbounded below if $A + \lambda I \not\succeq 0$ or if $A + \lambda I \succeq 0$ and $b \notin \mathcal{R}(A + \lambda I)$
- minimized by $x = -(A + \lambda I)^\dagger b$ otherwise: $g(\lambda) = -b^T (A + \lambda I)^\dagger b - \lambda$

dual problem and equivalent SDP:

$$\begin{array}{ll}\text{maximize} & -b^T (A + \lambda I)^\dagger b - \lambda \\ \text{subject to} & A + \lambda I \succeq 0 \\ & b \in \mathcal{R}(A + \lambda I)\end{array}$$

$$\begin{array}{ll}\text{maximize} & -t - \lambda \\ \text{subject to} & \begin{bmatrix} A + \lambda I & b \\ b^T & t \end{bmatrix} \succeq 0\end{array}$$

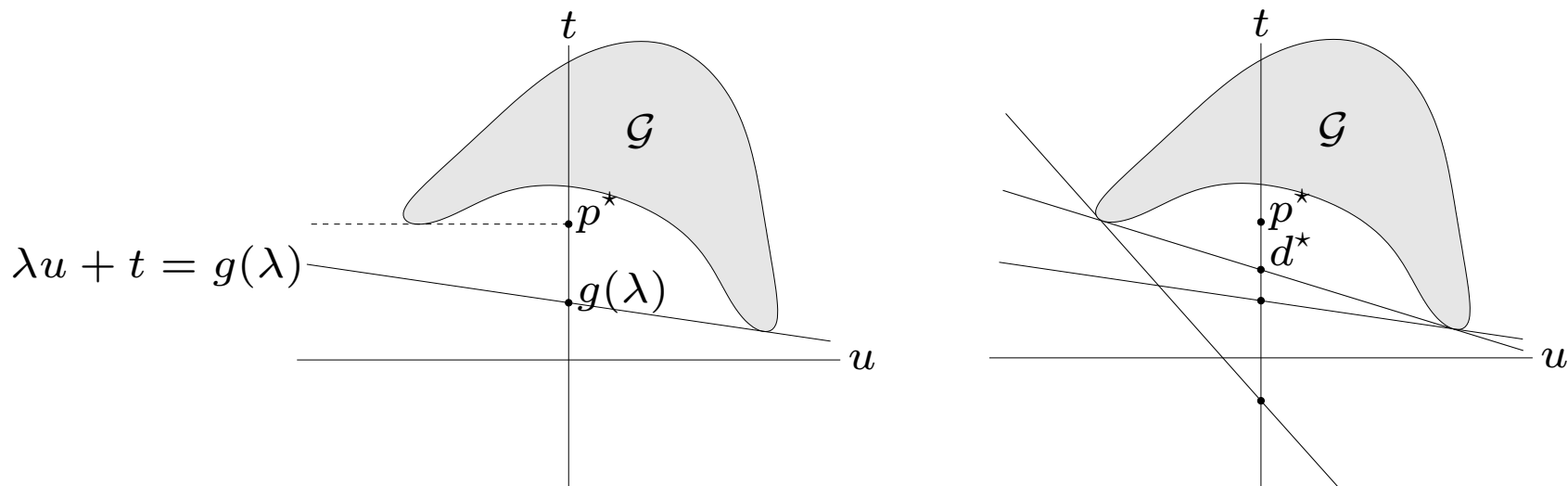
strong duality although primal problem is not convex (not easy to show)

Geometric interpretation

for simplicity, consider problem with one constraint $f_1(x) \leq 0$

interpretation of dual function:

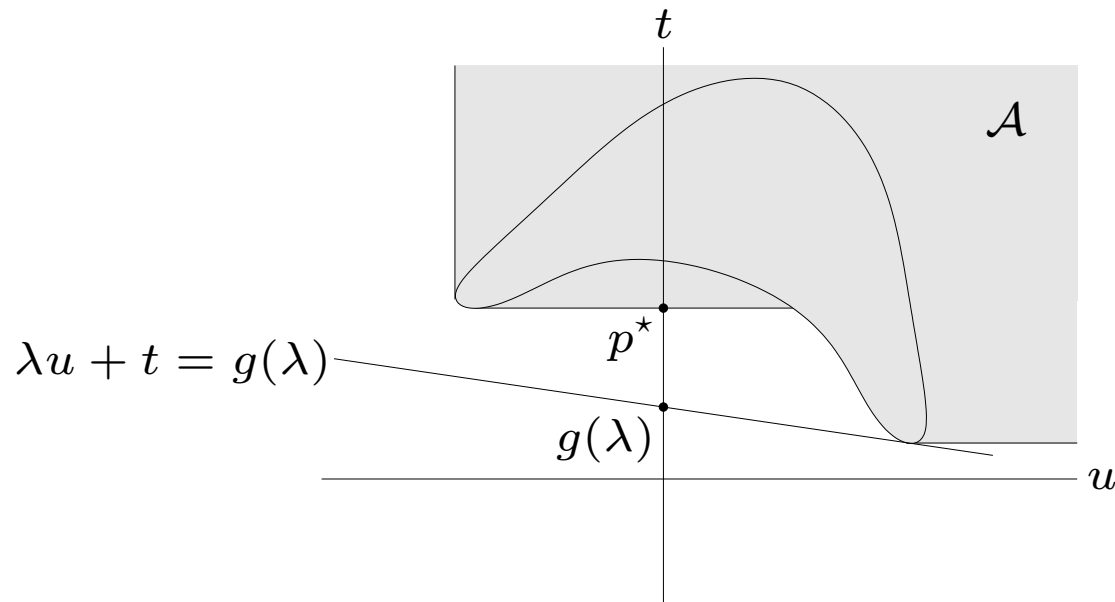
$$g(\lambda) = \inf_{(u,t) \in \mathcal{G}} (t + \lambda u), \quad \text{where } \mathcal{G} = \{(f_1(x), f_0(x)) \mid x \in \mathcal{D}\}$$



- $\lambda u + t = g(\lambda)$ is (non-vertical) supporting hyperplane to $\mathcal{G}$
- hyperplane intersects t -axis at $t = g(\lambda)$

epigraph variation: same interpretation if $\mathcal{G}$ is replaced with

$$\mathcal{A} = \{(u, t) \mid f_1(x) \leq u, f_0(x) \leq t \text{ for some } x \in \mathcal{D}\}$$



strong duality

- holds if there is a non-vertical supporting hyperplane to $\mathcal{A}$ at $(0, p^*)$
- for convex problem, $\mathcal{A}$ is convex, hence has supp. hyperplane at $(0, p^*)$
- Slater's condition: if there exist $(\tilde{u}, \tilde{t}) \in \mathcal{A}$ with $\tilde{u} < 0$, then supporting hyperplanes at $(0, p^*)$ must be non-vertical

Complementary slackness

assume strong duality holds, x^* is primal optimal, (λ^*, ν^*) is dual optimal

$$\begin{aligned} f_0(x^*) = g(\lambda^*, \nu^*) &= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{i=1}^p \nu_i^* h_i(x) \right) \\ &\leq f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{i=1}^p \nu_i^* h_i(x^*) \\ &\leq f_0(x^*) \end{aligned}$$

hence, the two inequalities hold with equality

- x^* minimizes $L(x, \lambda^*, \nu^*)$
- $\lambda_i^* f_i(x^*) = 0$ for $i = 1, \dots, m$ (known as complementary slackness):

$$\lambda_i^* > 0 \implies f_i(x^*) = 0, \quad f_i(x^*) < 0 \implies \lambda_i^* = 0$$

Karush-Kuhn-Tucker (KKT) conditions

the following four conditions are called KKT conditions (for a problem with differentiable f_i, h_i):

1. primal constraints: $f_i(x) \leq 0, i = 1, \dots, m, h_i(x) = 0, i = 1, \dots, p$
2. dual constraints: $\lambda \succeq 0$
3. complementary slackness: $\lambda_i f_i(x) = 0, i = 1, \dots, m$
4. gradient of Lagrangian with respect to x vanishes:

$$\nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + \sum_{i=1}^p \nu_i \nabla h_i(x) = 0$$

from page 5–17: if strong duality holds and x, λ, ν are optimal, then they must satisfy the KKT conditions

KKT conditions for convex problem

if $\tilde{x}$, $\tilde{\lambda}$, $\tilde{\nu}$ satisfy KKT for a convex problem, then they are optimal:

- from complementary slackness: $f_0(\tilde{x}) = L(\tilde{x}, \tilde{\lambda}, \tilde{\nu})$
- from 4th condition (and convexity): $g(\tilde{\lambda}, \tilde{\nu}) = L(\tilde{x}, \tilde{\lambda}, \tilde{\nu})$

hence, $f_0(\tilde{x}) = g(\tilde{\lambda}, \tilde{\nu})$

if **Slater's condition** is satisfied:

x is optimal if and only if there exist λ , ν that satisfy KKT conditions

- recall that Slater implies strong duality, and dual optimum is attained
- generalizes optimality condition $\nabla f_0(x) = 0$ for unconstrained problem

KKT Condition: Sufficiency and Necessity

Note: Slater's condition + Convex problem $\Rightarrow$ Strong duality + Dual optimum attained (if finite).

Optimality conditions

- Strong duality + attained at $(x^*, \lambda^*, \nu^*) \Rightarrow$ complementary slackness;
- Necessity of KKT conditions (for differentiable functions):
 - Strong duality + attained at $(x^*, \lambda^*, \nu^*) \Rightarrow$ KKT conditions;
 - In other words: KKT conditions are necessary if: strong duality + attained at (x^*, λ^*, ν^*) + differentiable.
- Sufficiency of KKT conditions (for differentiable functions):
 - KKT conditions $((\tilde{x}, \tilde{\lambda}, \tilde{\nu})$ solve KKT) + convex problem $\Rightarrow$ must an optimal point;
 - In other words: KKT conditions are sufficient if: convex + differentiable.
- Convex + Slater's condition (and finite optimum) $\Rightarrow$ KKT conditions are both necessary and sufficient for optimality.

Exercise 2

Consider the problem

$$\begin{aligned} &\text{minimize: } e^{-x} \\ &\text{subject to: } x^2/y \leq 0, \end{aligned}$$

with variables (x, y) and domain $\{(x, y) | y > 0\}$.

1. Is this a convex problem? What is the optimal value?
2. Find the Lagrange dual, and the optimal solution λ^* .
3. Does strong duality hold?

Exercise 3

Consider the problem

$$\begin{array}{ll}\text{minimize:} & x \\ \text{subject to:} & x - 2 \leq 0.\end{array}$$

1. Is this a convex problem? What is the optimal value?
2. Find the Lagrange dual, and the optimal value.
3. Does strong duality hold?
4. Find the KKT condition. Are the KKT conditions necessary in this case?

example: water-filling (assume $\alpha_i > 0$)

$$\begin{array}{ll} \text{minimize} & -\sum_{i=1}^n \log(x_i + \alpha_i) \\ \text{subject to} & x \succeq 0, \quad \mathbf{1}^T x = 1 \end{array}$$

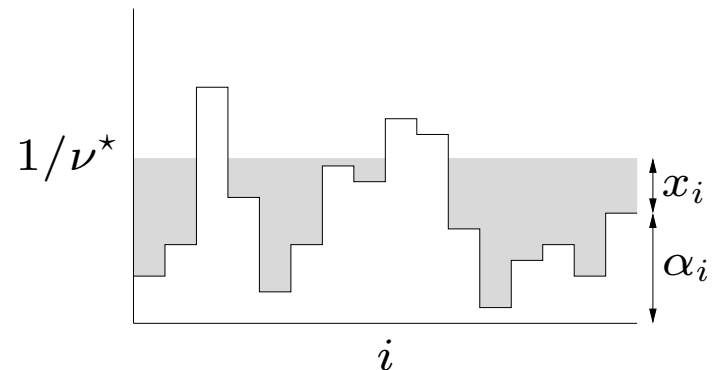
x is optimal iff $x \succeq 0$, $\mathbf{1}^T x = 1$, and there exist $\lambda \in \mathbf{R}^n$, $\nu \in \mathbf{R}$ such that

$$\lambda \succeq 0, \quad \lambda_i x_i = 0, \quad \frac{1}{x_i + \alpha_i} + \lambda_i = \nu$$

- if $\nu < 1/\alpha_i$: $\lambda_i = 0$ and $x_i = 1/\nu - \alpha_i$
- if $\nu \geq 1/\alpha_i$: $\lambda_i = \nu - 1/\alpha_i$ and $x_i = 0$
- determine ν from $\mathbf{1}^T x = \sum_{i=1}^n \max\{0, 1/\nu - \alpha_i\} = 1$

interpretation

- n patches; level of patch i is at height α_i
- flood area with unit amount of water
- resulting level is $1/\nu^*$



Perturbation and sensitivity analysis

(unperturbed) optimization problem and its dual

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

$$\begin{array}{ll}\text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0\end{array}$$

perturbed problem and its dual

$$\begin{array}{ll}\text{min.} & f_0(x) \\ \text{s.t.} & f_i(x) \leq u_i, \quad i = 1, \dots, m \\ & h_i(x) = v_i, \quad i = 1, \dots, p\end{array}$$

$$\begin{array}{ll}\text{max.} & g(\lambda, \nu) - u^T \lambda - v^T \nu \\ \text{s.t.} & \lambda \succeq 0\end{array}$$

- x is primal variable; u, v are parameters
- $p^*(u, v)$ is optimal value as a function of u, v
- we are interested in information about $p^*(u, v)$ that we can obtain from the solution of the unperturbed problem and its dual

global sensitivity result

assume strong duality holds for unperturbed problem, and that λ^* , ν^* are dual optimal for unperturbed problem

apply weak duality to perturbed problem:

$$\begin{aligned} p^*(u, v) &\geq g(\lambda^*, \nu^*) - u^T \lambda^* - v^T \nu^* \\ &= p^*(0, 0) - u^T \lambda^* - v^T \nu^* \end{aligned}$$

sensitivity interpretation

- if λ_i^* large: p^* increases greatly if we tighten constraint i ($u_i < 0$)
- if λ_i^* small: p^* does not decrease much if we loosen constraint i ($u_i > 0$)
- if ν_i^* large and positive: p^* increases greatly if we take $v_i < 0$;
if ν_i^* large and negative: p^* increases greatly if we take $v_i > 0$
- if ν_i^* small and positive: p^* does not decrease much if we take $v_i > 0$;
if ν_i^* small and negative: p^* does not decrease much if we take $v_i < 0$

local sensitivity: if (in addition) $p^*(u, v)$ is differentiable at $(0, 0)$, then

$$\lambda_i^* = -\frac{\partial p^*(0, 0)}{\partial u_i}, \quad \nu_i^* = -\frac{\partial p^*(0, 0)}{\partial v_i}$$

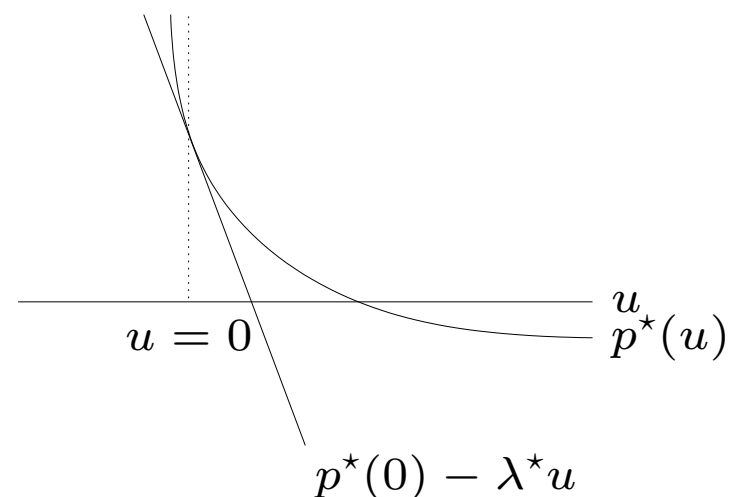
proof (for λ_i^*): from global sensitivity result,

$$\frac{\partial p^*(0, 0)}{\partial u_i} = \lim_{t \searrow 0} \frac{p^*(te_i, 0) - p^*(0, 0)}{t} \geq -\lambda_i^*$$

$$\frac{\partial p^*(0, 0)}{\partial u_i} = \lim_{t \nearrow 0} \frac{p^*(te_i, 0) - p^*(0, 0)}{t} \leq -\lambda_i^*$$

hence, equality

$p^*(u)$ for a problem with one (inequality)
constraint:



Duality and problem reformulations

- equivalent formulations of a problem can lead to very different duals
- reformulating the primal problem can be useful when the dual is difficult to derive, or uninteresting

common reformulations

- introduce new variables and equality constraints
- make explicit constraints implicit or vice-versa
- transform objective or constraint functions
e.g., replace $f_0(x)$ by $\phi(f_0(x))$ with ϕ convex, increasing

Introducing new variables and equality constraints

$$\text{minimize } f_0(Ax + b)$$

- dual function is constant: $g = \inf_x L(x) = \inf_x f_0(Ax + b) = p^*$
- we have strong duality, but dual is quite useless

reformulated problem and its dual

$$\begin{array}{ll} \text{minimize} & f_0(y) \\ \text{subject to} & Ax + b - y = 0 \end{array}$$

$$\begin{array}{ll} \text{maximize} & b^T \nu - f_0^*(\nu) \\ \text{subject to} & A^T \nu = 0 \end{array}$$

dual function follows from

$$\begin{aligned} g(\nu) &= \inf_{x,y} (f_0(y) - \nu^T y + \nu^T Ax + b^T \nu) \\ &= \begin{cases} -f_0^*(\nu) + b^T \nu & A^T \nu = 0 \\ -\infty & \text{otherwise} \end{cases} \end{aligned}$$

norm approximation problem: minimize $\|Ax - b\|$

$$\begin{array}{ll}\text{minimize} & \|y\| \\ \text{subject to} & y = Ax - b\end{array}$$

can look up conjugate of $\|\cdot\|$, or derive dual directly

$$\begin{aligned}g(\nu) &= \inf_{x,y} (\|y\| + \nu^T y - \nu^T Ax + b^T \nu) \\ &= \begin{cases} b^T \nu + \inf_y (\|y\| + \nu^T y) & A^T \nu = 0 \\ -\infty & \text{otherwise} \end{cases} \\ &= \begin{cases} b^T \nu & A^T \nu = 0, \quad \|\nu\|_* \leq 1 \\ -\infty & \text{otherwise} \end{cases}\end{aligned}$$

(see page 5–4)

dual of norm approximation problem

$$\begin{array}{ll}\text{maximize} & b^T \nu \\ \text{subject to} & A^T \nu = 0, \quad \|\nu\|_* \leq 1\end{array}$$

Implicit constraints

LP with box constraints: primal and dual problem

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & -\mathbf{1} \preceq x \preceq \mathbf{1} \end{array} \qquad \begin{array}{ll} \text{maximize} & -b^T \nu - \mathbf{1}^T \lambda_1 - \mathbf{1}^T \lambda_2 \\ \text{subject to} & c + A^T \nu + \lambda_1 - \lambda_2 = 0 \\ & \lambda_1 \succeq 0, \quad \lambda_2 \succeq 0 \end{array}$$

reformulation with box constraints made implicit

$$\begin{array}{ll} \text{minimize} & f_0(x) = \begin{cases} c^T x & -\mathbf{1} \preceq x \preceq \mathbf{1} \\ \infty & \text{otherwise} \end{cases} \\ \text{subject to} & Ax = b \end{array}$$

dual function

$$\begin{aligned} g(\nu) &= \inf_{-\mathbf{1} \preceq x \preceq \mathbf{1}} (c^T x + \nu^T (Ax - b)) \\ &= -b^T \nu - \|A^T \nu + c\|_1 \end{aligned}$$

dual problem: maximize $-b^T \nu - \|A^T \nu + c\|_1$

Exercise 4

The general form LP is:

$$\begin{aligned} &\text{minimize: } c^T x \\ &\text{subject to: } Gx \preceq h \\ &\quad \quad \quad Ax = b. \end{aligned}$$

1. Find the dual function.
2. Find the dual problem, and rewrite it by making the implicit equality constraints explicit.
3. Compare it with the dual of the standard form LP.
4. Compare it with the dual of the inequality form LP.

Exercise 4: Solution for question (3)

The general form LP is:

$$\begin{aligned} &\text{minimize: } c^T x \\ &\text{subject to: } Gx \preceq h \\ &\quad \quad \quad Ax = b. \end{aligned}$$

Dual LP:

$$\begin{aligned} &\text{maximize: } -h^T \lambda - b^T \nu \\ &\text{subject to: } c + G^T \lambda + A^T \nu = 0 \\ &\quad \quad \quad \lambda \succeq 0. \end{aligned}$$

Standard form LP:

$$\begin{aligned} &\text{minimize: } c^T x \\ &\text{subject to: } Ax = b \\ &\quad \quad \quad x \succeq 0. \end{aligned}$$

Dual LP:

$$\begin{aligned} &\text{maximize: } -b^T \nu \\ &\text{subject to: } A^T \nu - \lambda + c = 0 \\ &\quad \quad \quad \lambda \succeq 0. \end{aligned}$$

Equivalent dual LP:

$$\begin{aligned} &\text{maximize: } -b^T \nu \\ &\text{subject to: } A^T \nu + c \succeq 0. \end{aligned}$$

Exercise 4: Solution for question (4)

The general form LP is:

$$\begin{aligned} &\text{minimize: } c^T x \\ &\text{subject to: } Gx \preceq h \\ &\quad Ax = b. \end{aligned}$$

Dual LP:

$$\begin{aligned} &\text{maximize: } -h^T \lambda - b^T \nu \\ &\text{subject to: } c + G^T \lambda + A^T \nu = 0 \\ &\quad \lambda \succeq 0. \end{aligned}$$

Inequality form LP:

$$\begin{aligned} &\text{minimize: } c^T x \\ &\text{subject to: } Ax \preceq b. \end{aligned}$$

Dual LP:

$$\begin{aligned} &\text{maximize: } -b^T \lambda \\ &\text{subject to: } A^T \lambda + c = 0 \\ &\quad \lambda \succeq 0. \end{aligned}$$

Exercise 5

Recall the Boolean linear program: p^*

minimize: $c^T x$

subject to: $Ax \preceq b$

$$x_i \in \{0, 1\}, \quad i = 1, 2, \dots, n.$$

1. Consider the optimization problem,

minimize: $c^T x$

subject to: $Ax \preceq b$

$$x_i(1 - x_i) = 0, \quad i = 1, 2, \dots, n.$$

Is this convex? Find its dual, which is another lower bound for p^* .

2. Which lower bound for p^* is better? (This question is optional: they are the same.)

Related to an LP: a lower bound for p^*

minimize: $c^T x$

subject to: $Ax \preceq b$

$$0 \leq x_i \leq 1, \quad i = 1, 2, \dots, n.$$

Exercise 6

Consider the problem

$$\begin{aligned} &\text{minimize: } x_1^2 + x_2^2 \\ &\text{subject to: } (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ &\quad \quad \quad (x_1 - 1)^2 + (x_2 + 1)^2 \leq 2. \end{aligned}$$

1. What is the feasible set and the optimal solution?
2. Derive the KKT conditions. Can you find a solution? What does it mean?